INCH-POUND

MIL-I-82847(OS)
5 February 1990
SUPERSEDING
WS 22370B
9 December 1986

MILITARY SPECIFICATION

IGNITION COMPOUND, MAGNESIUM-POLYTETRAFLUOROETHYLENE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers one type of magnesium-polytetrafluoroethylene ignition compound, referred to herein as the compound.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116 Preservation, Methods of

MIL-M-382 Magnesium Powder (For Use in Ammunition)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1337

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MIL-P-82810 Polytetrafluoroethylene (PTFE)

STANDARDS

FEDERAL

FED-STD-102 Preservation, Packaging, and Packing Levels

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Character or quality.** All raw materials used in the compound shall be tested and accepted in accordance with the following:

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a. Magnesium shall be in accordance with MIL-M-382, Type III, Granulation Number 16.

b. Polytetrafluoroethylene shall be in accordance with MIl-P-82810.

3.2 **Formulation.** Formulation of the compound shall be in accordance with table I. The compound shall be a completely blended mixture of the ingredients.

TABLE I. Formulation.

Constituent Material	Percent by Weight
Magnesium	60 ± 3
Polytetrafluoroethylene	40 ± 3

3.3 **Chemical and physical properties.** Chemical and physical properties of the compound shall be in accordance with table II.

TABLE II. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Moisture content, percent	---	0.5
Heat of reaction, cal/g	1300	---
Autoignition, (°F) (for not less than 2 hours)	350	---

3.4 **Shelf life.** The compound shall have a shelf life of 5 years when stored in the original, desiccated, sealed containers at ambient conditions not exceeding 110°F.

3.5 **Safety.**

3.5.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.5.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

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3.6 **Workmanship.** Workmanship shall be such that the compound is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table III.

TABLE III. Quality conformance inspection.

Examination and Tests	Requirement	Test Method
Visual examination	3.1, 3.6, and Section 5	4.3.1
Composition	3.2	4.3.1.1
Formulation	3.2	4.3.2
Moisture content	3.3	4.3.3
Heat of reaction	3.3	4.3.4
Autoignition	3.3	4.3.5

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4.2.1 **Lot.** A lot shall consist of all the compound submitted for inspection at one time from one of the following:

a. A single batch (see 6.3.1).

b. A continuous process (see 6.3.2).

c. A uniform blend of batches or a uniform blend or portions from a continuous process.

4.2.2 **Sampling.** The samples shall be selected at random in the quantity specified in table IV. The selected single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites except where a lot consists of fewer than three containers. A single determination of each quality characteristic listed in table III shall be performed on each of the three composite samples independently. When a lot consists of fewer than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall meet the requirements of table II. Failure of any composite sample to meet any requirement of table II shall be cause for rejection of the lot.

TABLE IV. Sampling.

Number of containers in Lot	Number of Containers to be Selected
1 - 3	All
4 - 15	3
16 - 25	4
26 - 40	5
41 - 65	7
66 - 110	10
111 - 180	15

4.2.3 **Classification of characteristics.** The characteristics verified by the inspections herein are classified as Major or Minor in accordance with WR-43. Inspections that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. Inspections not annotated with a classification code and serial number are classified minor.

4.3 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the test or procedure description, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

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4.3.1 **Visual examination.** Material acceptance records and contractor certification shall be visually examined to verify conformance of raw materials to 3.1 and 3.2 (see 6.2). The compound shall be visually examined for conformance to 3.6 and section 5.

4.3.1.1 **Composition.** The composition of the compound, measured as percent polytetrafluoroethylene, shall be determined by carefully reacting a sample of 2.0 ± 0.4 grams of well-blended compound with 10 percent hydrochloric acid, adding a drop or two of a surface-active agent (surfactant) (Basic H, etc.), refluxing the unreacted residue, rinsing with distilled water, filtering, drying the residue and weighing to determine the percent polytetrafluoroethylene.

4.3.2 **Formulation.** Weigh-up sheets and contractor certification for each batch shall be examined to verify that formulation used in the compound lot is in accordance with 3.2 (see 6.2).

4.3.3 **Moisture content (M101).** The moisture content shall be determined by weight-loss after conditioning at 110 ± 5°C for not less than 14 hours.

4.3.4 **Heat of reaction (M102).** The heat of reaction of the compound shall be determined using a Parr Bomb Calorimeter, Model 1221, or equivalent, using the manufacturer's Manual Number 120, except that no water shall be added to the bomb and the test shall be conducted at 25 atmospheres of helium or argon after three purges of the bomb with helium or argon at that pressure.

4.3.5 **Autoignition.** A sample of 1.0 ± 0.2 grams of the compound shall not ignite when exposed to a temperature of 177 to 182°C (350 to 360°F) for 2 hours, minimum. The sample shall be placed in a container preconditioned to the required temperature and maintained at the temperature for the required time.

5. PACKAGING

5.1 **Preservation and packaging.** Levels of preservation and packaging shall be in accordance with FED-STD-102, Level B. The material shall be packaged in 1 pound increments in accordance with MIL-P-116, Method IId, except that no cushioning is required and the barrier material shall be conductive (see 6.4). The packaging shall also comply with 49 CFR 171-178.

5.2 **Packing.** When unit packages are multipacked for shipment, the level of packing shall be in accordance with FED-STD-102, Level B and shall not exceed 100 pounds of material.

5.3 **Marking.**

5.3.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.2 **Special marking.** In addition to the marking requirements of 5.3.1, each container marking shall include the following:

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a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Shelf life expiration date.

e. Storage conditions.

f. Toxicity warning (if applicable).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The compound is intended for use in the manufacture of rocket motor igniters for the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Material Safety Data Sheets required (see 3.7.1 and 6.6).

d. Inspection sampling plan (see 4.2.2).

e. DOD FAR 52.246-15 (see 4.3.1 and 4.3.2).

f. Special marking, if other than as specified in 5.3.2.

g. Safety precautions required (see 6.6).

6.3 Definitions.

6.3.1 Batch. A batch is that amount of compound made as one unit in the manufacturing process.

6.3.2 Continuous process. A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the compound is accumulated for a lot.

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6.4 **Conductive barrier material.** The conductive barrier material used for packaging shall be "Velostat" as manufactured by the 3M Company, St. Paul, MN 55144, or equivalent.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1337-N060)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-82847	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
IGNITION COMPOUND, MAGNESIUM-POLYTETRAFLUOROETHYLENE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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